

Causal Inference and Learned Helplessness

Anonymous CogSci submission

Abstract

Prolonged exposure to uncontrollable situations can cause individuals to become and remain dysfunctionally passive. This pattern, known as *learned helplessness*, is associated with depression, making understanding and mitigating it of clinical interest. Learned helplessness has been induced in lab settings using simple tasks, yet real-world control often involves complex, non-linear causal systems where the ability to influence an outcome diverges from the ease with which one can bring about the outcome one *wants*, and where ascriptions of self-agency are non-trivial, depending on prior mechanistic beliefs and counterfactual inference. We examine learned helplessness in dynamic control environments, allowing participants to interact with causal variables as they influence one another in real time, while we systematically manipulate structure, controllability and reward prevalence. Whilst low levels of practical control reliably induced helpless behaviour, we found this did not depend on reward prevalence or the accuracy of learners' causal models.

Keywords: causal inference; learned helplessness; control; counterfactuals

Learned helplessness is a psychological phenomenon in which individuals, after experiencing uncontrollable adverse events, come to expect that their actions will have little or no effect on future outcomes. This expectation transfers to new situations, leading to pervasive and dysfunctional passivity. The phenomenon was first documented in animal work (Overmier & Seligman, 1967) where dogs receiving unavoidable electrical shocks later did not take action to escape even when escape was possible. This pattern has been demonstrated in human adults in analogous tasks involving inescapable loud tones (Hiroto, 1974), and shown to depend on the extent participants causally attribute outcomes to external factors. Learned helplessness is commonly used as a model of depression, since experimentally-induced helplessness replicates depressed individuals' diminished perception of control (Miller & Seligman, 1975).

Pathological control perception is hypothesised to play a central role in learned helplessness, where participants fail to recognise contingency between their actions and outcomes (Maier & Seligman, 1976). Subsequent findings challenge the accuracy and utility of this account. The depressive realism hypothesis (Alloy & Abramson, 1979) suggests depressed individuals make more accurate judgments about action-outcome contingencies than non-depressed individuals, who tend to overestimate their control. Indeed, neurotypical individuals exhibit optimism bias (Sharot, 2011) and il-

lusions of control, particularly when desired outcomes are common (Langer, 1975; Langer & Rodin, 1976). Teodorescu and Erev (2014) found that prevalence of rewards in an experimental task played a larger role in preventing helpless behaviour than actual action-outcome contingencies.

Nevertheless, other work emphasises controllability as the key factor in learned helplessness. The opportunity to choose, even in arbitrary decisions, increases motivation and engagement (Cordova & Lepper, 1996). Choice is preferred over non-choice, and experienced as rewarding, as demonstrated across self-report, behavioural, and neuroimaging data (Leotti & Delgado, 2011). Rewards have little effect on performance in induction or problem-solving tasks (Osborn Popp et al., 2025). This suggests that where identifying action-outcome relations is key, participants may be motivated by an intrinsic desire for discovery and control. In interactive gamified tasks, curiosity and signals of learning progress, may be stronger determinants of task performance than absolute rewards (Chu & Schulz, 2020).

Humans construct causal representations of their environment to facilitate reasoning, learning, and decision-making, particularly in contexts that require prediction, explanation or adaptive control (Sloman, 2005). Much distinctively human cognition can be construed as a “causal model-based control”, where success depends on learning and exploiting a causal model to achieve changing goals (Broadbent et al., 1986; Davis et al., 2018; Schulz et al., 2017). Learned helplessness scenarios fit this mould: participants must learn at test that a button press terminates an electric shock and use this knowledge to escape harm. Thus, this phenomenon can be understood as either failure to construct a causal model that includes this affordance at test time, or failure to leverage that causal model to plan and perform an action that influence outcomes.

Several parallels make this framing appealing: Causal strength judgments (e.g., Cheng, 1997) build on the the same kinds of instrumental contingency utilised by Maier and Seligman (1976); research on attribution theory, which focuses on inferring the causes of social behaviour (e.g., Kelley, 1973), has informed later developments in learned helplessness theory which involve attributional styles (Abramson et al., 1978). While recent studies have explored causal model-based control in continuous time with continuous variables (Davis et al., 2018), the influence of individual differences and experimental manipulations on adaptive behaviour remain under-

studied. Furthermore, learned helplessness paradigms have largely relied on static, atemporal designs that fail to capture the complexity of real-world contexts that involve continuous functional relationships, delays, and feedback loops, where the timing and sequence of events are crucial for causal inference and effective control (Cartwright, 2004; Sloman & Lagnado, 2015).

The present study addresses a gap in the literature by investigating the role of causal inference in learned helplessness. We examine learned helplessness within a dynamic control task—an interactive online game where participants engage with continuous variables across discrete time-points. Adopting an exploratory approach that synthesises research on causal reasoning and dynamic decision-making, we manipulate features of the training phase to test two key predictions: that lower objective controllability will increase helpless behaviour, in line with prior work on perceived contingency, and that lower reward prevalence will similarly increase helplessness, consistent with findings on the illusion of control.

Methods

Framework: Ornstein-Uhlenbeck process

To create causal environments where participants interact with continuous variables in discrete time, following (Davis et al., 2020), we use an Ornstein-Uhlenbeck (OU) process to define how a set of continuous variables interact with each other across time steps. This process characterises a variable’s random movement in time, biased towards a mean μ (Uhlenbeck & Ornstein, 1930). The change in variable x , Δx_t for a time step t is defined as:

$$\Delta x_t = \omega[\mu - x_t] + N(0, \sigma) \quad (1)$$

where $\omega > 0$ is a value determining how strongly attracted x is to μ . The value of x is continuously perturbed with normally distributed noise of variance σ . The result is a “mean regressive random walk”, that is, a variable which evolves randomly over time but with a basin of attraction centred on μ .

This framework can be extended to capture causal dynamics involving multiple variables by making the attraction states of variables nonstationary, and defining them as functions of their causal parents. To take the case of a three-variable causal system x , y , and z connected by a possibly cyclic causal graph G , the constant μ is replaced at time $t + 1$ with a function of the values of the causal parents at time t (Davis et al., 2020). The evolution of this system is then described by a system of equations of the form:

$$\Delta x_{t+1} = \omega \left[\left(\sum_{y \in a(x)} \theta_t^{yx} y_t \right) - x_t \right] + N(0, \sigma) \quad (2)$$

where $a(x)$ are the parents of x in the causal graph. Here, x tends towards a new value determined by a linear combination of causal variables y and z , which are modified by θ representing the existence and strength of causal relationships

$y \rightarrow x$ and $z \rightarrow x$. If $\theta_{yx} = 0$, y is not causally related to x ; if $\theta_{yx} > 0$, x tends towards a multiple of y ’s value (it is attracted); if $\theta_{yx} < 0$, x tends towards a negative multiple of y ’s value (it is repelled). Importantly, this relationship is independent for both directions, such that $\theta_{yx} = 0$ does not imply $\theta_{xy} = 0$. Cyclic relationships between two variables influencing each other such that $\theta_{yx} \neq 0$ and $\theta_{xy} \neq 0$ are also possible. If x has no parents, the system assumes it is attracted to 0.

Using this framework, we systematically vary the amount of control participants can exert over the system. For high control conditions, we followed (Davis et al., 2020) in setting θ at $[1, -1, 0]$ for “regular”, “inverse”, and “none” causal relationships. This enabled participants to more easily assess causal strength and relationships. σ was set to 4 as a baseline level of noise. For low control conditions, we drew $\theta_t \sim \text{Beta}(1, 3)$ independently on each step for regular connections and multiplied it with -1 for inverse connections. This meant that causes would often have little influence on their effects but occasionally have sizeable impact limiting participants ability to exert fine-grained prospective control over the system, as their actions had unpredictable effects. We also used $\sigma = 18$ to obscure the causal relationships by injecting a larger amount of exogenous noise into the system. All participants had stable $[1, -1, 0]$ θ values and $\sigma = 4$ during the test phase. Figure 1a illustrates the different behaviour of each causal structure across the control manipulations.

We used six different causal structures across the experimental rounds, each consisting of three variables x , y , z labelled “X”, “Y”, and “Z” (Figure 1b). The variables changed value in discrete time steps according to Equation 2. The structures were randomly counterbalanced within subject by swapping X and Y (Figure 1b). Participants were presented with a different structure in each round, requiring them to discover the underlying causal mechanisms while generalising task-level knowledge across rounds. All structures were rated as equally difficult by participants in a prior study. They were also designed so that influencing Z could be achieved through intervention on just one of X or Y.

Experiment

Preregistration can be accessed at <https://aspredicted.org/vf52-wxvr.pdf>, and the data and analysis code, and videos of the task are hosted on the Open Science Framework at https://osf.io/2yx8t/overview?view_only=bb9c86a736164cfc8a39359863cf4cfc. Ethical approval for this study was granted by *anonymised for peer review*. A power analysis detailed on OSF suggested minimum sample of 77. To ensure adequate power after potential exclusions, we recruited 100 participants via Prolific and compensated £2.00 each ($M_{\text{time}} = 19.7$; $SD_{\text{time}} = 5.3$). Following our preregistered criteria, seven participants were excluded for failing comprehension questions more than twice, or disengaging by switching browser tab, yielding a final sample of 93 ($M_{\text{age}} = 43.3$; $SD_{\text{age}} = 13.0$; $N_{\text{female}} = 48$ [51.6%]).

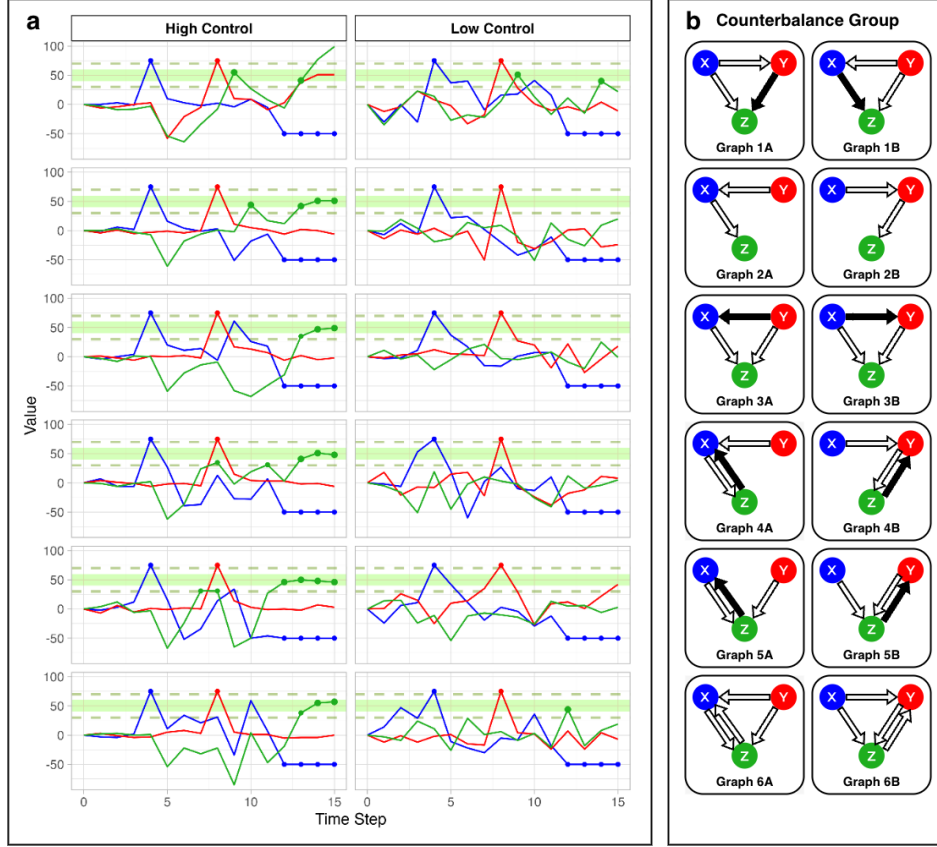


Figure 1: (a) Behaviour of causal structures in counterbalance group A when perturbed with the same sequence of interventions (at timesteps 4, 8, 12, 13, 14, & 15), across both high and low control conditions. The small reward region is shaded in green, and the large reward region is marked with dashed lines. (b) Causal structures counterbalanced between groups A and B. Black arrows represent regular relationships ($\theta = 1$), and white arrows represent inverse relationships ($\theta = -1$).

Round-level procedure Each round, participants were presented with a task interface (Figure 2) where the three variables within the causal structure are displayed as separate sliders labelled X, Y, and Z. These were not related to any real world variables, in order to reduce the influence of prior knowledge that could inflate or deflate participants’ perceived control (Btsh et al., 2025; Yarritu & Matute, 2015). The variables shifted their values in discrete time across thirty 2.5 second timesteps, according to the transition process defined in Equation 2 conditioned on any participant actions. The total score and current timestep was displayed at the top of the interface. The trial ended after $30 \times 2.5 = 75$ seconds.

Participants could interact with only one of sliders X or Y on each timestep, by clicking and dragging or holding the handle. This interaction allowed them to perform a control action, an intervention on variables X or Y, overriding the OU network dynamics and the variable’s causal parents (Pearl, 2000). By exerting control, participants could attempt to influence the system and bring the target variable Z into the reward region. These interventions could move the variables to any value between the range $[-100, 100]$. If a participant moved a slider during a timestep, the variable took the most

recent position set by the participant and remained at that value for the rest of the timestep. At the transition to the next timestep, any variable that had not been intervened on during the previous timestep was updated according to the OU dynamics. When participants held a slider across two or more timesteps, each timestep that the variable held counted as an individual intervention. Thus if slider Y was held across timesteps 9-12, we counted this as 4 separate interventions.¹

To introduce a cost-benefit dynamic, each intervention incurred a 1 point cost. Placing the target slider in the reward region yielded a larger reward of 5 points per timestep. Participants could choose not to intervene, but this would rarely result in the target slider reaching or remaining in the reward region. Thus, inaction was generally nonoptimal. This structure allows learned helplessness to manifest as disengagement from control which becomes dysfunctional when the expected immediate (or future) returns of acting are greater than zero: participants in low control conditions may stop interacting entirely after experiencing negative returns, resulting in no further rewards but also no loss of points, effectively main-

¹This diverges from Davis et al. (2020)’s characterisation where an intervention only ended once the participant released their cursor.

taining their current score.

A line chart displayed the values of each variable over the most recent 15 timesteps, allowing participants to track the history of their actions. The slider and chart elements were restricted to the $[-100, 100]$ range and variables pushed outside this range by the dynamics were truncated to stay within these limits. A shaded reward region was displayed, centred on $y = 50$ spanning $[30, 70]$ in the large reward region conditions and $[40, 60]$ in the small reward region conditions (Figure 1a).

After each round, participants were invited to complete an extra question. They were asked to indicate a causal link they thought was most likely to be present in the previous round. This question mandated the choice of one of the 6×2 potential causal edges ($X \rightarrow Y, \dots, Z \rightarrow X$), and aimed to probe which causal link participants had leveraged the most in their control strategy. We opted not to ask participants to make complete causal structure judgments to keep the experiment reasonably short and focused on the control task.

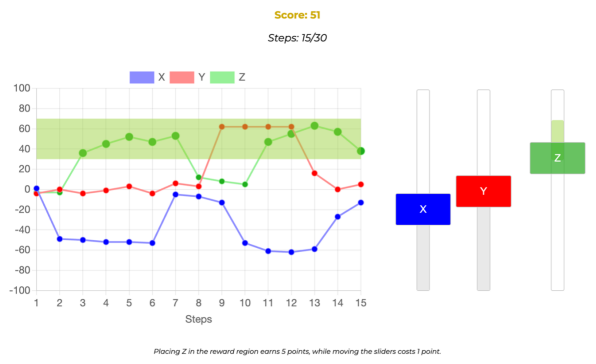


Figure 2: Task Interface screenshot from high control and high reward condition. The reward region is shown in green. Participants could set variable X or Y or do nothing, and their goal was to control variable Z. In this case the true causal structure is graph 2A (see Figure 1). The participant was rewarded on steps 3-7 and 11-15. There are 15 steps remaining in the round.

Design & Procedure We deployed the study online using JavaScript. After providing informed consent, participants received detailed instructions including videos and annotated diagrams to explain the experimental setup and task objectives. Participants then had to pass a comprehension quiz to proceed to the main task. Participants were randomly assigned to one of the four experimental conditions; causal structure presentation order was randomised and counterbalanced across rounds, ensuring participants had equal likelihood of encountering structures from groups A and B. The experimental conditions were varied in a 2×2 between-subject design. The size of the reward region (large vs. small) was crossed with the degree of control afforded by the dynamic causal system (high vs. low).

Participants completed three rounds each followed by a post-

round question during the training phase. After the training phase, participants viewed their total score and rated the task’s perceived controllability and difficulty on an 11-point Likert scale. They then completed a structurally identical test phase, with scores reset at the first test round. Critically, all participants had equal control during the test phase; participants who had low control during training gained control equal to high control conditions. The size of the reward region remained constant for each condition across training and test rounds. Following the test phase, participants again rated the task’s perceived controllability and difficulty.

We collected a range of behavioural measures such as perceived controllability, number of interventions, number of rewarded timesteps, and accuracy in identifying a causal relation. Additionally, we included the Hope Scale (Snyder et al., 1991) as a post-experiment measure. We selected this over depression inventories such as the BDI (Beck et al., 2011) due to the study’s focus on perceived controllability rather than clinical symptoms. This was followed by demographic questions (e.g., age, gender), input method (mouse vs. trackpad), and an open-ended question on their control strategy.

Results

Pre-registered analyses

There were no significant differences in interventions across the three test rounds ($\beta = -0.19$, $SE = 0.32$, $t(185) = -0.60$, $p = .551$), so we fit all subsequent models on collapsed data. We evaluated two linear mixed effects models: A full model including all covariates and a reduced model containing only the variables of interest (control and reward region). Formal likelihood ratio testing confirmed that adding additional predictors (age, trackpad use, self-report controllability, causal link identification, and Hope Scale) did not significantly improve model fit beyond the core variables ($\chi^2(5) = 1.02$, $p = .961$). We therefore present the reduced model.

Model predictions are visualised in Figure 3. Participants who had low control during their training phase performed fewer interventions compared to their high control counterparts ($\beta = -4.57$, $SE = 1.73$, $t = -2.65$, $p = .010$), when holding reward region size constant. Contrary to our prediction, the size of reward region had no significant association with intervention count ($\beta = -0.68$, $SE = 1.73$, $t = -0.39$, $p = .696$). We found no correlation between participant’s reported Hope scale and their number of interventions ($r(91) = -0.09$, $p = .404$).

Exploratory analyses

Task performance First, we used two-tailed Welch’s t -tests to assess if round-level scoring differed from no-control or random-control strategies. The no-control strategy involved making no interventions throughout the round, incurring no cost. The random-control strategy applied an intervention at each timestep to either X or Y, with the intervention value

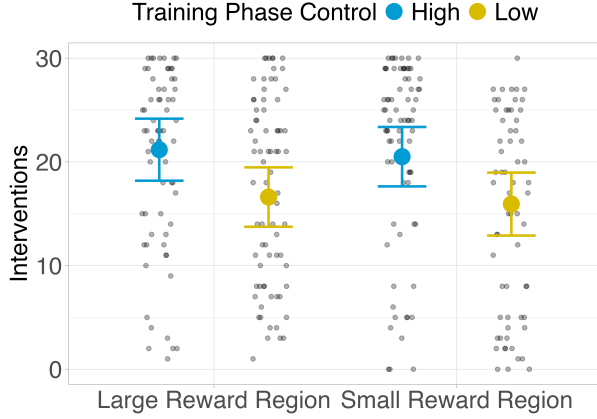


Figure 3: Model predicted intervention counts (with 95% confidence intervals) based on control and reward region size, superimposed over raw data from these conditions.

drawn uniformly from the range $[-100, 100]$. For each strategy, we simulated 1000 rounds for all 6 causal graphs within each of the 4 different experimental conditions.

During the test phase, all conditions performed better than random interventions at each timestep. For participants with low control: small reward region ($t(62.29) = 6.33, p < .001$); large reward region ($t(74.45) = 2.74, p = .008$). For participants with high control: small reward region ($t(74.14) = 6.39, p < .001$); large reward region ($t(65.18) = 3.85, p < .001$). When compared to no-control simulations, performance did not differ significantly for the low control–small reward region condition ($t(62.00) = 0.45, p = .654$), but was significantly higher for low control–large reward region ($t(74.03) = 3.23, p = .002$), high control–small reward region ($t(74.00) = 2.61, p = .011$), and high control–large reward region ($t(65.01) = 4.18, p < .001$) conditions.

Figure 4 shows within-round success by condition by timestep with linear model fits. A smaller proportion of participants succeeded per trial in the low control condition compared to the high control condition ($\beta = -0.04, SE = 0.02, t(114) = -2.22, p = .029$). Second, participants in the small reward region conditions showed similar decreases in scoring proportion ($\beta = -0.04, SE = 0.02, t(114) = -2.12, p = .036$). While the proportion of scoring participants increased by 1% each timestep ($SE = 0.009, t(114) = 10.69, p < .001$). There was a significant control $\times$ timestep interaction ($\beta = -0.002, SE = 0.0009, t(114) = -3.28, p < .001$) indicating that participants in low control conditions had impaired improvement trajectories. Whilst both control and reward prevalence independently affected performance, low control specifically hampered performance over time within rounds.

Perceived controllability We used cumulative link mixed-models (CLMM) from the Ordinal package in R (Chris-

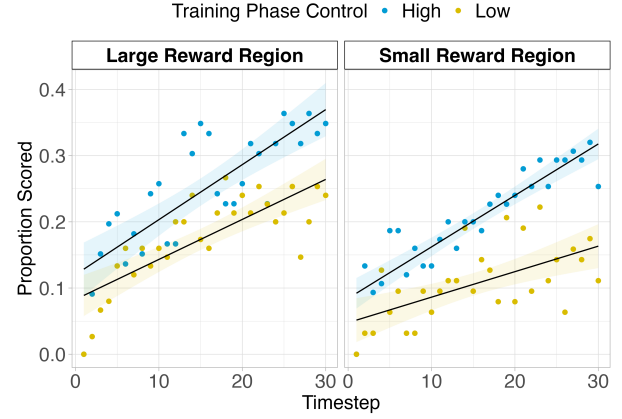


Figure 4: Proportion of participants scoring in each timestep during test trials, along with estimated linear model fits.

tensen, n.d.) to fit participants self-reported perceived controllability during the training and test phase at the 11 Likert levels (0 – 10) as a function of condition. Participants in the high control and large reward region conditions had higher odds, 3.8 (95%CI[1.4, 10], $p = .008$) and 2.8 (95%CI[1.0, 7.7], $p = .044$) respectively, of reporting higher ratings of perceived controllability.

Identifying causal links Participants were above chance ($>25\%$) in identifying a causal link correctly in both training ($n = 279, 95\%CI[0.28, 0.40], p = .001$) and test ($n = 279, 95\%CI[0.27, 0.39], p = .003$) phases. This performance was not uniform across all conditions in the test phase. Only participants with small reward regions achieved above-chance accuracy ($>25\%$) in identifying correct causal links post-test rounds, for high ($n = 75, 95\%CI[0.31, 0.55], p < .001$) and low control ($n = 63, 95\%CI[0.25, 0.50], p = .04$) subsets. Participants with large reward regions. Causal link identification accuracy of participants with large reward regions did not differ significantly from chance for both high ($n = 66, 95\%CI[0.13, 0.35], p = .777$) and low control ($n = 75, 95\%CI[0.19, 0.41], p = .423$). Surprisingly, a mixed-effects logistic regression also revealed that participants in small reward region conditions had twice the odds of correctly identifying causal links ($OR = 1.92, 95\%CI[1.10, 3.37], p = .022$) compared to participants with large reward regions. Amount of control during the training phase was not significant ($p = .872$) in predicting causal link identification.

Quality of interventions Finally, we simulated what would have occurred had participants not intervened at a given timestep, for each intervention performed in the task. An example can be found in Figure 5. If a participant intervened on variable X at timestep t, we used the values of its parent variables at t-1 within the causal structure to determine the pre-intervention value of X at t. Based on the current values at t, we then used Equation 2 to calculate the counterfactual value of Z at t+1 and determine whether it was within the

reward range. As in the experimental task, we used $\theta = [1, -1, 0]$ for high control conditions, whilst we used $\theta = [0.25, -0.25, 0]$ for low control conditions (expected value of the beta distribution). By comparing the counterfactual value of Z to the actual value of Z , interventions could be classified as detrimental (reward lost due to intervention), inconsequential (no effect on reward outcome), necessary (reward only achieved with intervention), or redundant (reward achieved regardless). Counts of intervention types performed during the training phase are shown in Table 1.

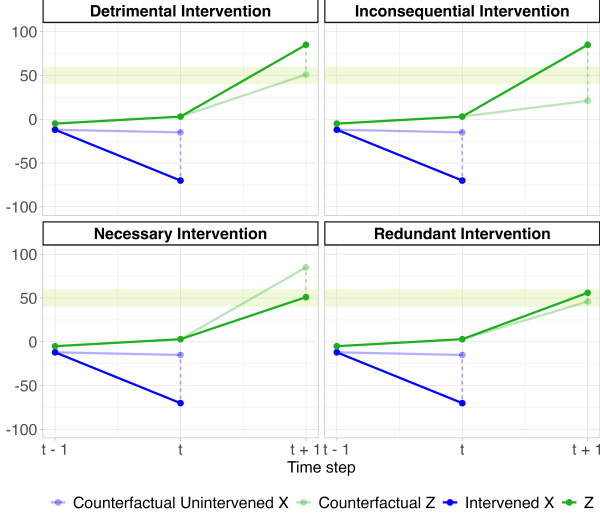


Figure 5: Visual explanation of different intervention types. In each panel, variable X was intervened on at t (dark blue). The counterfactual value of variable Z at $t+1$ (light green) is calculated based on the value variable X would have taken (light blue) without the intervention. Values of variable X (a timestep $t+1$) and Y (at all timesteps) have been omitted for clarity.

We found that some intervention types were significant predictors of test-phase interventions. Specifically, a higher number of inconsequential ($\beta = 0.03, SE = 0.00, t(88) = 7.54, p < .001$), necessary ($\beta = 0.04, SE = 0.01, t(88) = 3.99, p < .001$), and redundant ($\beta = 0.06, SE = 0.03, t(88) = 2.18, p = .032$) interventions were positively associated with the number of test-phase interventions. The number of detrimental interventions in training did not significantly predict test-phase interventions ($p = .127$). Moreover, a greater number of necessary ($OR = 1.15, 95\%CI[1.09, 1.21], p < .001$) and redundant ($OR = 1.16, 95\%CI[1.04, 1.29], p = .008$) interventions was associated with higher perceived controllability. In contrast, inconsequential interventions were negatively associated with controllability ratings ($OR = 0.96, 95\%CI[0.93, 0.98], p < .001$).

Discussion

This study offers a novel exploration of learned helplessness in a dynamic control setting. We found that participants who

Table 1: Count of intervention types in the training phase by condition

Condition	Detri.	Incon.	Neces.	Redun.
High Reward High Control	90	810	248	94
Low Reward High Control	68	1140	50	196
High Reward Low Control	0	1356	156	0
Low Reward Low Control	0	1071	53	0

experienced uncontrollable training rounds interacted less in a test phase, and performed worse than those in high control conditions. Whilst reward prevalence did not impact helplessness measures, participants in the low reward conditions were more accurate in identifying causal links in their respective causal graphs. One interpretation is the low reward conditions necessitated attention to causal relations, whereas in the high reward conditions, participants could rely on chance or model-free strategies like win-stay-lose-sample (Robbins, 1952). This also supports the notion that when tasks require inductive reasoning, intrinsic motivations such as curiosity and perceived agency may outweigh extrinsic rewards in guiding behaviour (Leotti et al., 2010; Osborn Popp et al., 2025).

The dynamic nature of our experimental framework offers a distinct methodological advantage over static paradigms: the ability to quantify the precise efficacy of every action. This analysis revealed a dissociation between the feeling of control and the act of controlling. While the number of necessary, redundant, and even inconsequential interventions during training predicted more test-phase interventions, performing more inconsequential interventions reduced perceived controllability. The observed persistence in performing interventions, particularly involving costly actions, aligns with the optimism bias often observed in non-clinical samples. However, it also suggests an intrinsic epistemic drive. Even in dynamic environments where low control implies noisy rather than zero contingency, participants may employ interventions to explore the system, attempting to distinguish causal signal from noise to build accurate causal models. Extending this fine-grained analysis to clinical contexts offers a path to transcend simple measures of passivity, by helping to pinpoint whether helpless behaviour stems from a lack of motivation or an inability to disentangle causal efficacy from noise.

Learned helplessness manifests in various aspects of life and often extends beyond well-defined, discrete experimental trials. Our study demonstrates that helpless behaviour can be induced within a dynamic causal environment, where structure, controllability and reward exert distinct influences at different stages of the process. This framework offers a promising first step in exploring not only learned helplessness but also a range of goal-directed adaptive behaviour within psychology. Grounded in causal representations, this approach opens new avenues for understanding how individuals navigate complex, changing environments.

References

- Abramson, L. Y., Seligman, M. E., & Teasdale, J. D. (1978). Learned helplessness in humans: Critique and reformulation. *Journal of Abnormal Psychology*, 87(1), 49–74. <https://doi.org/10.1037/0021-843X.87.1.49>
- Alloy, L. B., & Abramson, L. Y. (1979). Judgment of contingency in depressed and nondepressed students: Sadder but wiser? *Journal of Experimental Psychology: General*, 108(4), 441–485. <https://doi.org/10.1037/0096-3445.108.4.441>
- Beck, A. T., Ward, C. H., Mendelson, M., Mock, J., & Erbauch, J. (2011, September). Beck Depression Inventory. <https://doi.org/10.1037/t00741-000>
- Broadbent, D. E., FitzGerald, P., & Broadbent, M. H. P. (1986). Implicit and explicit knowledge in the control of complex systems. *British Journal of Psychology*, 77(1), 33–50. <https://doi.org/10.1111/j.2044-8295.1986.tb01979.x>
- Btsh, V., Bramley, N. R., Speekenbrink, M., & Lagnado, D. A. (2025). Less is more: Local focus in continuous time causal learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. <https://doi.org/10.1037/xlm0001451>
- Cartwright, N. (2004). Causation: One word, many things. *Philosophy of Science*, 71(5), 805–819. <https://doi.org/10.1086/426771>
- Cheng, P. W. (1997). From covariation to causation: A causal power theory. *Psychological Review*, 104(2), 367–405. <https://doi.org/10.1037/0033-295X.104.2.367>
- Christensen, R. H. B. (n.d.). Cumulative Link Models for Ordinal Regression with the R Package ordinal.
- Chu, J., & Schulz, L. E. (2020). Play, curiosity, and cognition. *Annual Review of Developmental Psychology*, 2(Volume 2, 2020), 317–343. <https://doi.org/10.1146/annurev-devpsych-070120-014806>
- Cordova, D. I., & Lepper, M. R. (1996). Intrinsic motivation and the process of learning: Beneficial effects of contextualization, personalization, and choice. *Journal of Educational Psychology*, 88(4), 715–730. <https://doi.org/10.1037/0022-0663.88.4.715>
- Davis, Z. J., Bramley, N. R., & Rehder, B. (2020). Causal structure learning in continuous systems. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.00244>
- Davis, Z. J., Bramley, N. R., Rehder, B., & Gureckis, T. (2018). A casual model approach to dynamic control. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 40(0).
- Hiroto, D. S. (1974). Locus of control and learned helplessness. *Journal of Experimental Psychology*, 102(2), 187–193. <https://doi.org/10.1037/h0035910>
- Kelley, H. H. (1973). The processes of causal attribution. *American Psychologist*, 28(2), 107–128. <https://doi.org/10.1037/h0034225>
- Langer, E. J. (1975). The illusion of control. *Journal of Personality and Social Psychology*, 32(2), 311–328. <https://doi.org/10.1037/0022-3514.32.2.311>
- Langer, E. J., & Rodin, J. (1976). The effects of choice and enhanced personal responsibility for the aged: A field experiment in an institutional setting. *Journal of Personality and Social Psychology*, 34(2), 191–198. <https://doi.org/10.1037/0022-3514.34.2.191>
- Leotti, L. A., & Delgado, M. R. (2011). The inherent reward of choice. *Psychological Science*, 22(10), 1310–1318. <https://doi.org/10.1177/0956797611417005>
- Leotti, L. A., Iyengar, S. S., & Ochsner, K. N. (2010). Born to choose: The origins and value of the need for control. *Trends in Cognitive Sciences*, 14(10), 457–463. <https://doi.org/10.1016/j.tics.2010.08.001>
- Maier, S. F., & Seligman, M. E. (1976). Learned helplessness: Theory and evidence. *Journal of Experimental Psychology: General*, 105(1), 3–46. <https://doi.org/10.1037/0096-3445.105.1.3>
- Miller, W. R., & Seligman, M. E. (1975). Depression and learned helplessness in man. *Journal of Abnormal Psychology*, 84(3), 228–238. <https://doi.org/10.1037/h0076720>
- Osborn Popp, P. J., Newell, B. R., Bartels, D. M., & Gureckis, T. M. (2025). Can cognitive discovery be incentivized with money? *Journal of Experimental Psychology: General*, 154(4), 1025–1037. <https://doi.org/10.1037/xge0001682>
- Overmier, J. B., & Seligman, M. E. (1967). Effects of inescapable shock upon subsequent escape and avoidance responding. *Journal of Comparative and Physiological Psychology*, 63(1), 28–33. <https://doi.org/10.1037/h0024166>
- Pearl, J. (2000). *Causality: Models, reasoning, and inference*. Cambridge University Press.
- Robbins, H. (1952). Some aspects of the sequential design of experiments. *Bulletin of the American Mathematical Society*, 58(5).
- Schulz, E., Klenke, E. D., Bramley, N. R., & Speekenbrink, M. (2017). Strategic exploration in human adaptive control. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 39(0).
- Sharot, T. (2011). The optimism bias. *Current Biology*, 21(23), R941–R945. <https://doi.org/10.1016/j.cub.2011.10.030>
- Sloman, S. (2005, August). *Causal Models: How People Think about the World and Its Alternatives* (1st ed.). Oxford University Press New York. <https://doi.org/10.1093/acprof:oso/9780195183115.001.0001>
- Sloman, S., & Lagnado, D. (2015). Causality in thought. *Annual Review of Psychology*, 66(1), 223–247. <https://doi.org/10.1146/annurev-psych-010814-015135>
- Snyder, C. R., Harris, C., Anderson, J. R., Holleran, S. A., Irving, L. M., Sigmon, S. T., Yoshinobu, L., Gibb, J., Langelle, C., & Harney, P. (1991). The will and the ways: Development and validation of an individual-differences measure of hope. *Journal of Personality and Social Psychology*, 60(4), 570–585. <https://doi.org/10.1037/0022-3514.60.4.570>
- Teodorescu, K., & Erev, I. (2014). Learned helplessness and learned prevalence: Exploring the causal relations among perceived controllability, reward prevalence, and explo-

- ration. *Psychological Science*, 25(10), 1861–1869. <https://doi.org/10.1177/0956797614543022>
- Uhlenbeck, G. E., & Ornstein, L. S. (1930). On the theory of the Brownian Motion. *Physical Review*, 36(5), 823–841. <https://doi.org/10.1103/PhysRev.36.823>
- Yarritu, I., & Matute, H. (2015). Previous knowledge can induce an illusion of causality through actively biasing behavior. *Frontiers in Psychology*, 6. <https://doi.org/10.3389/fpsyg.2015.00389>